Diameter Growth Equations for CONUS with Environment and Climate Variables

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2026-11-05

Preliminary CONUS-wide Diameter Growth Equation

Data and Model Form

We built a data set from Forest Inventory and Analysis (FIA) remeasurement data, filtering out observations with missing values for diameter at breast height (`dbh`), basal area in larger trees (`bal`), crown ratio (`cr`), total height (`ht`), site index (`si`), and remeasurement intervals less than 5 years. We also restricted the data to species with ≥ 5000 observations.

The first two principal components of environmental/climatic variables are used. They are:

$$pc_1 = 0.252DD_0 + 0.002TD - 0.035PPT_{sm} + 0.967DD_{18} \tag{1}$$

$$pc_2 = 0.882DD_0 + 0.015TD + 0.42PPT_{sm} - 0.215DD_{18} \tag{2}$$

where:

- DD_0 = growing degree days above $0^{\circ}C$
- DD_{18} = growing degree days above $18^{\circ}C$
- TD = Temperature Difference (difference between the mean temperature of the warmest month (MWMT) and the mean temperature of the coldest month (MCMT)
- PPT_{sm} = summer precipitation (mm)

The diameter growth equation fit to these data is:

$$\Delta dbh = e^{(\beta_0+\beta_1 log(\frac{(dbh+1)^{\beta_3}}{(cr*ht+1)^{\beta_4}})+\beta_2 \frac{bal^{\beta_5}}{dbh+2.7}+\beta_5 pc_1+\beta_6 pc_2} \tag{3}$$

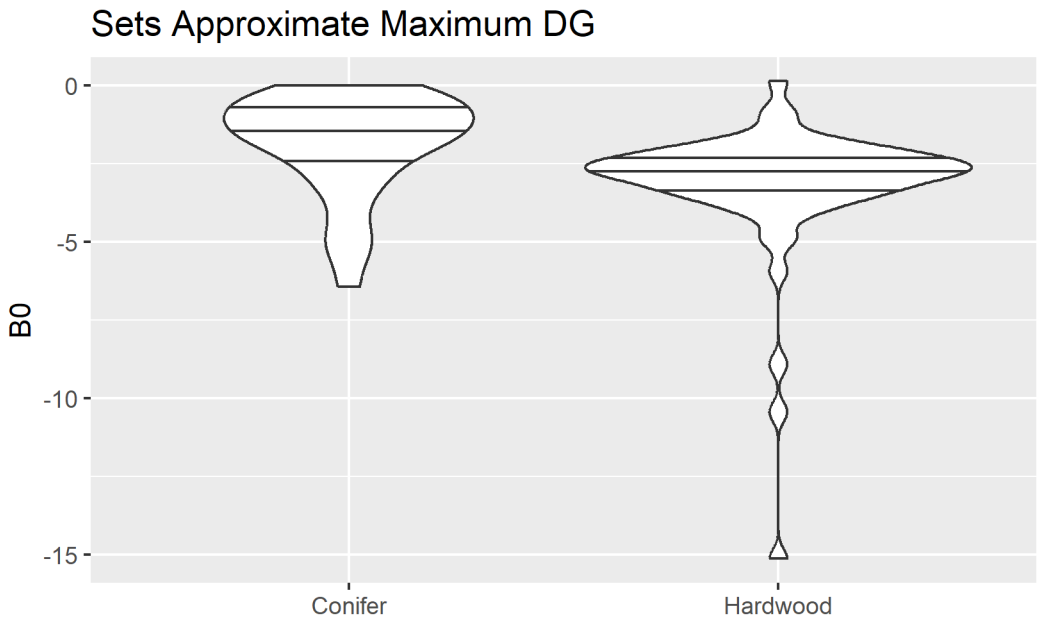
where:

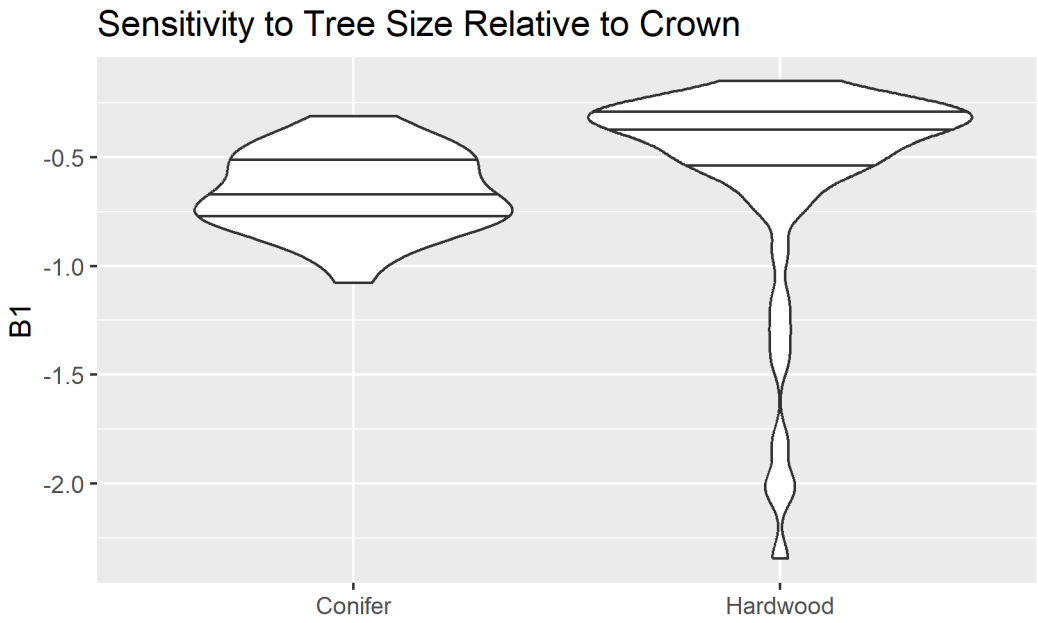
- dbh = diameter at breast height (inches)
- cr = crown ratio (fraction of total height)
- ht = total height (feet)
- bal = basal area in larger trees ($feet^2/acre$)
- pc_1 = environmental/climate principal component 1
- pc_2 = environmental/climate principal component 2

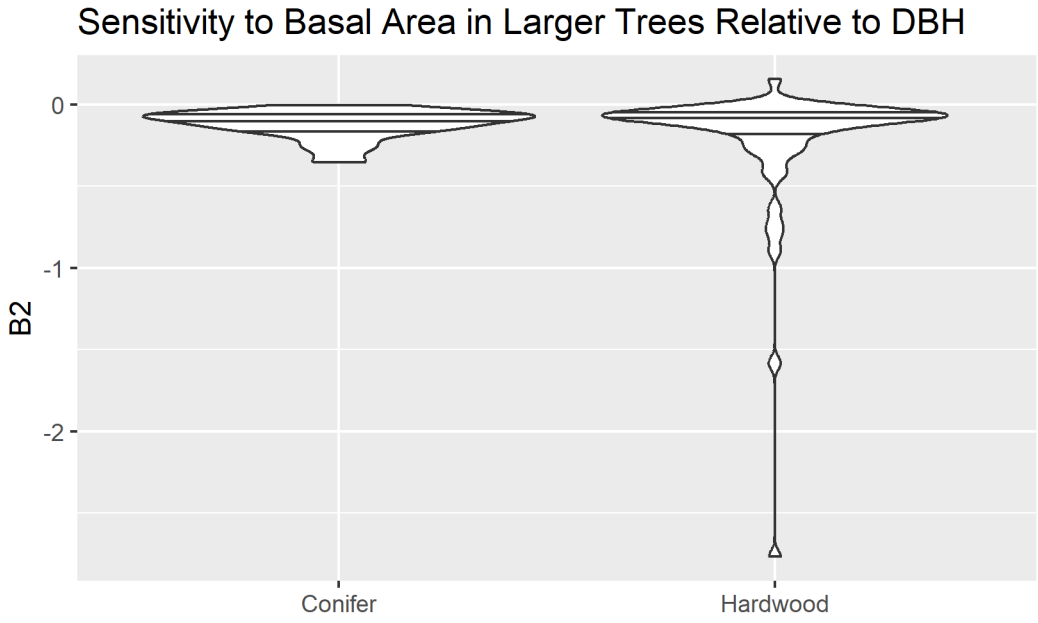
Equation 3 was fit to each qualifying species using an integrated fitting approach and errors minimized were ending `dbh`. 89 species equations were estimated. Sample sizes ranged from 5148 to 301494 observations.

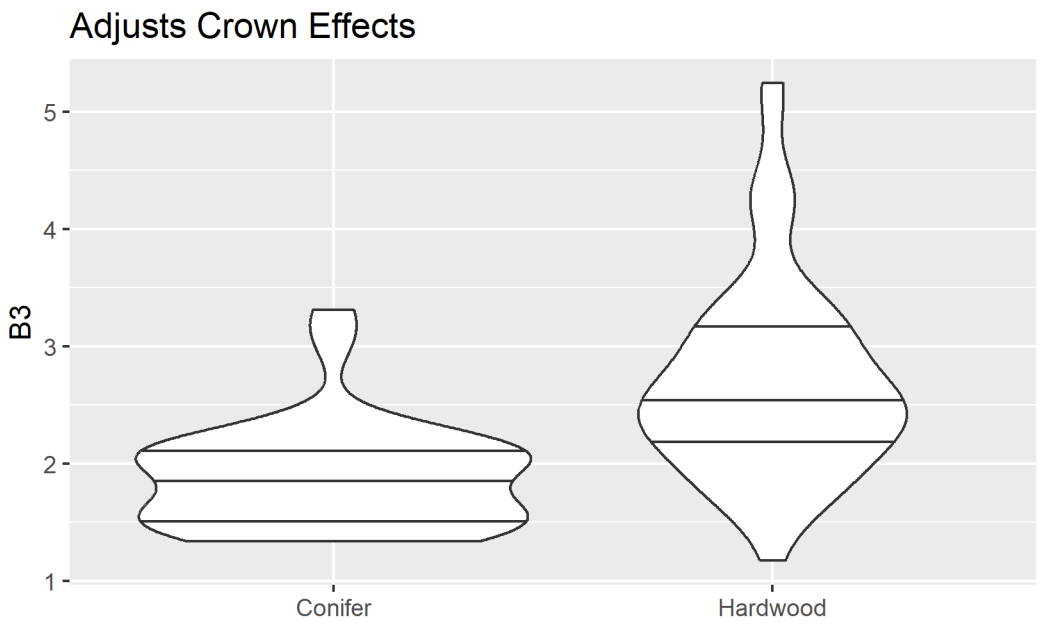
Fitting Results

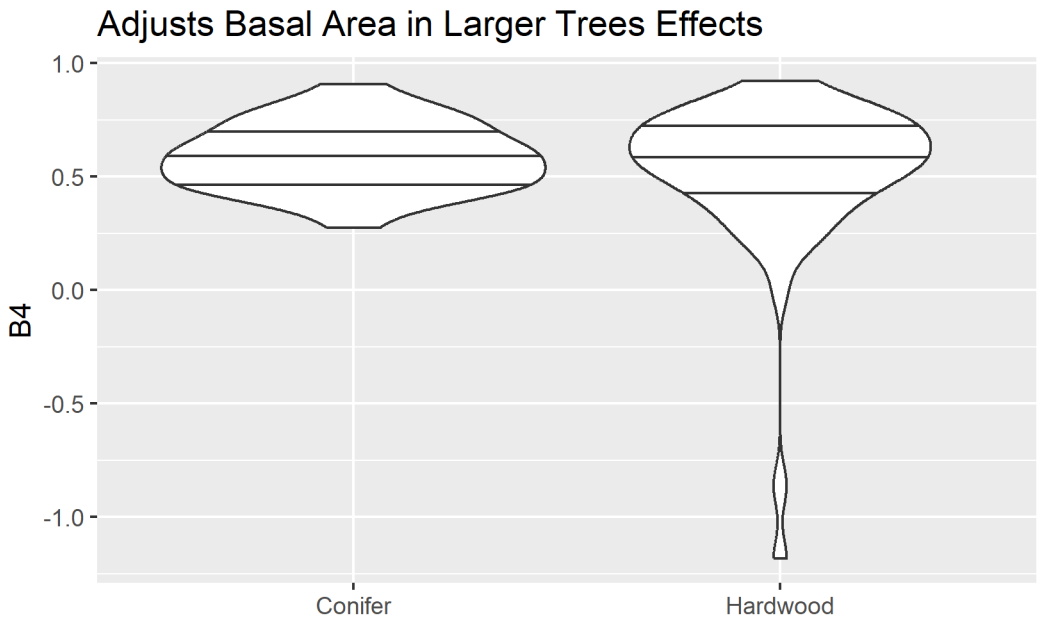
The graphs below show the range of parameter estimates divided into Conifer and Hardwood groups. The parameter estimates are remarkable stable across species.

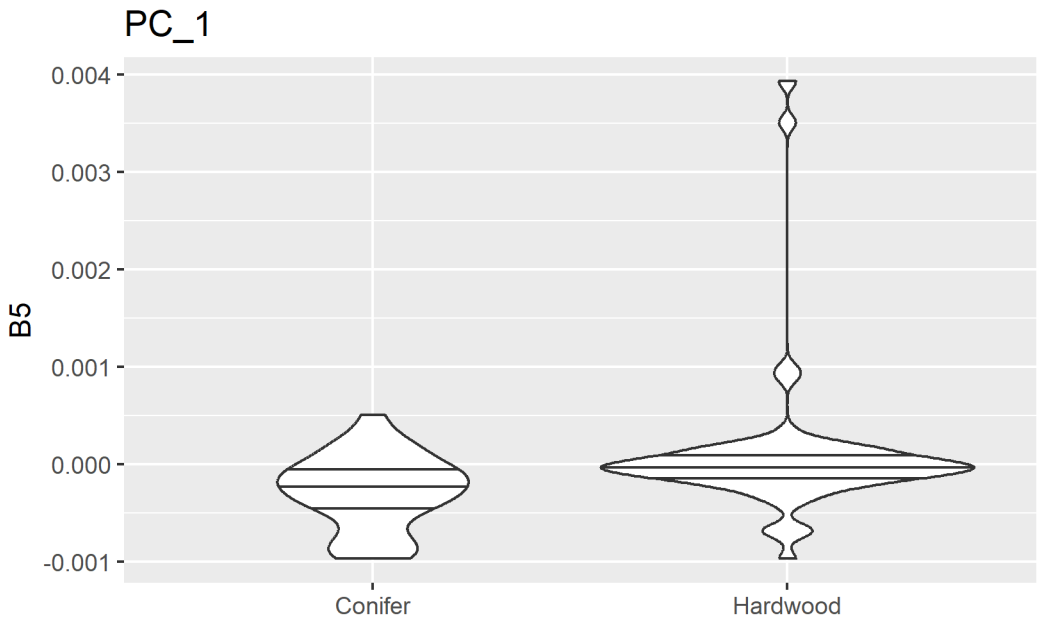



Sensitivity to Tree Size Relative to Crown
B1
-0.5
-1.0
-1.5
-2.0
Conifer
Hardwood

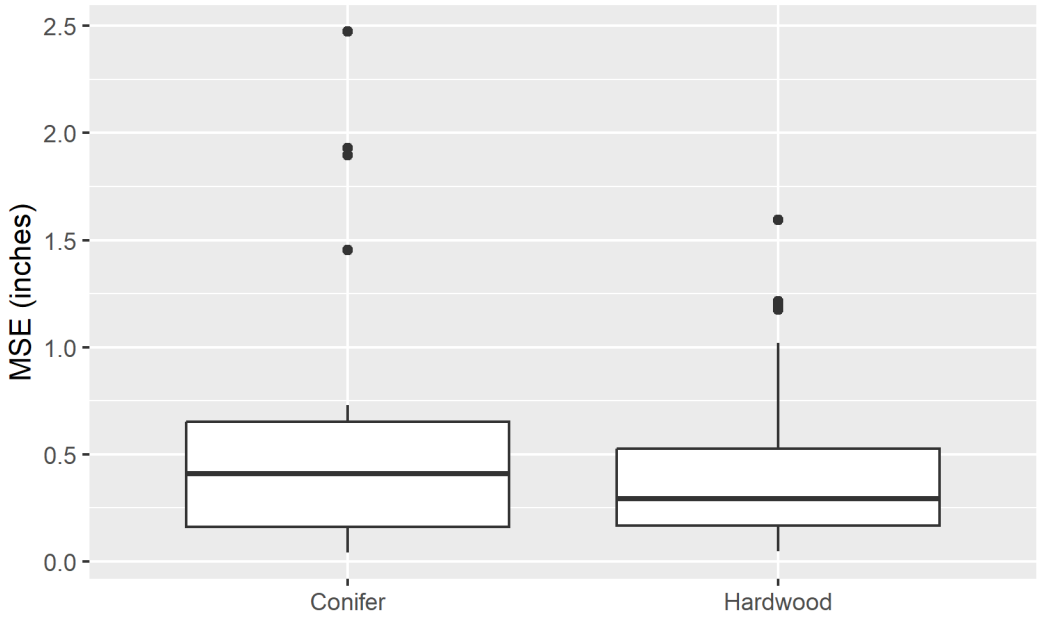

Sensitivity to Basal Area in Larger Trees Relative to DBH
B2
0
-1
-2
Conifer
Hardwood


Adjusts Crown Effects
B3
5
4
3
2
1
Conifer
Hardwood

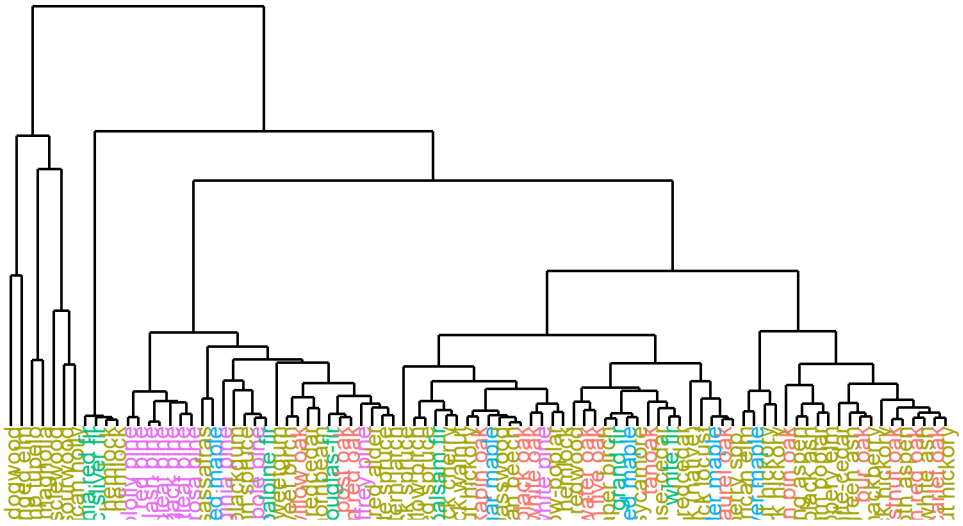

Adjusts Basal Area in Larger Trees Effects
B4
1.0
0.5
0.0
-0.5
-1.0
Conifer
Hardwood


PC_1
B5
0.004
0.003
0.002
0.001
0.000
-0.001
Conifer
Hardwood

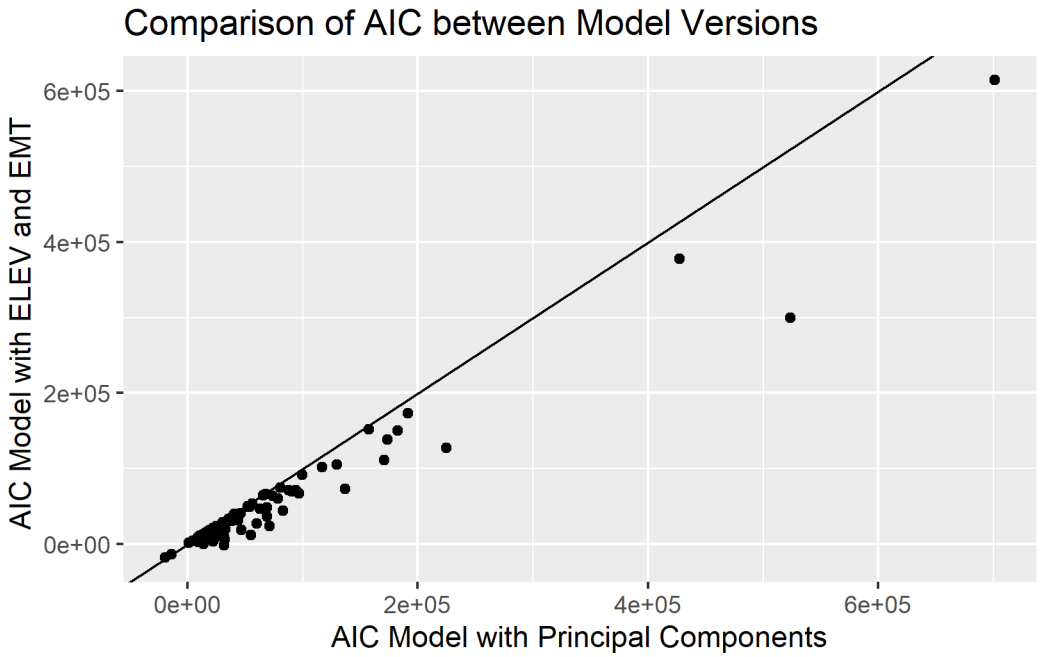

PC_2
B6
0.03
0.02
0.01
0.00
-0.01
Conifer
Hardwood



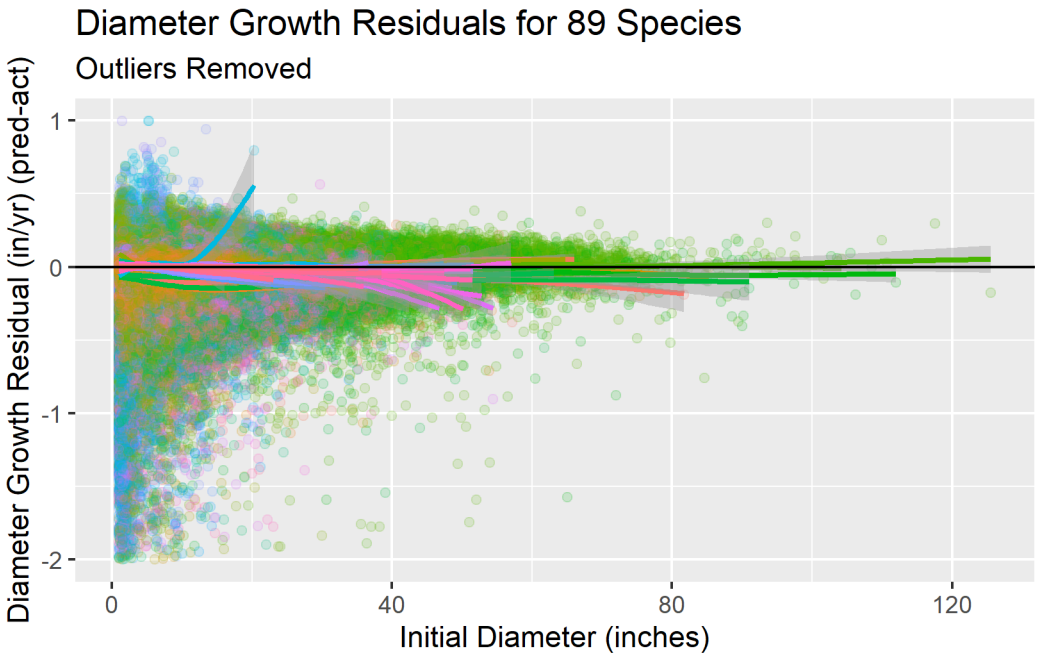
FIA Species Clusters by Diameter Growth Parameter Estimates

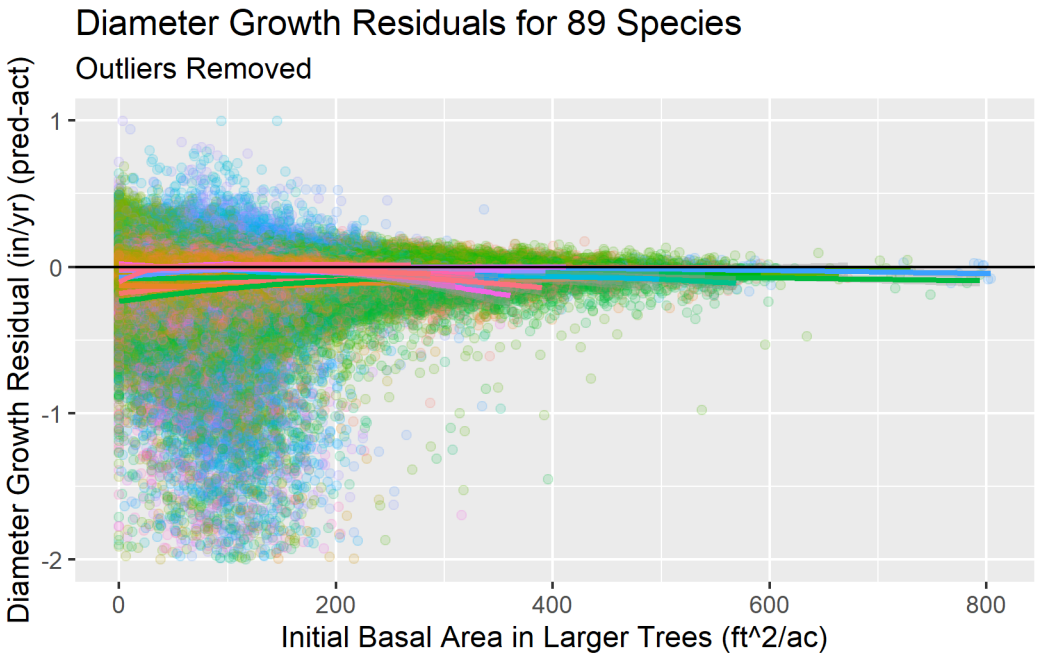


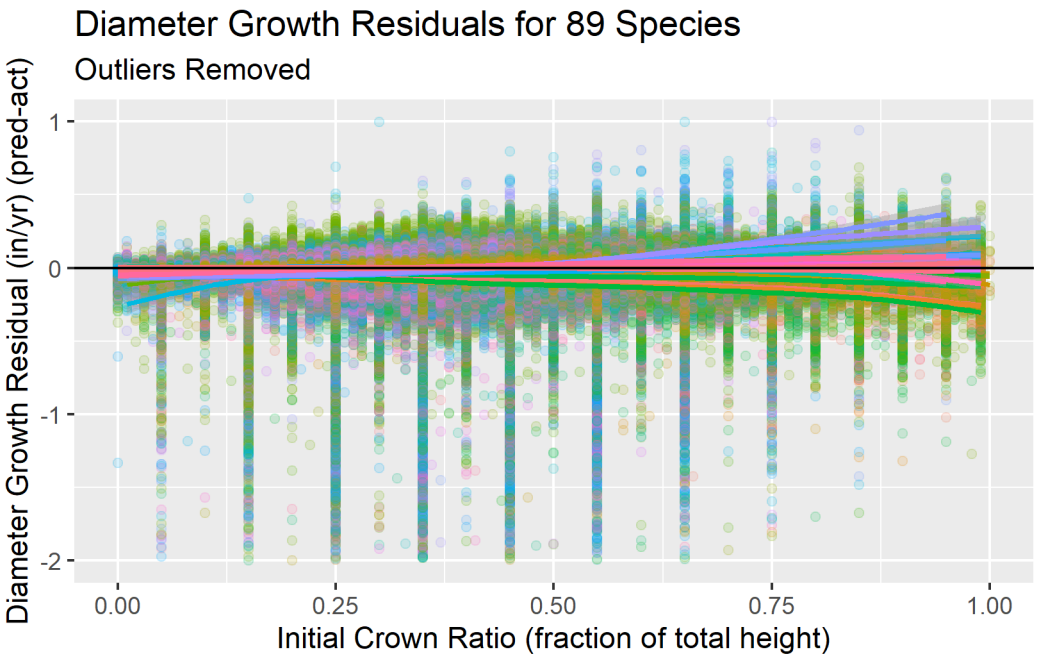
Comparison with Previous Model using Elevation and Extreme Minimum Temperature

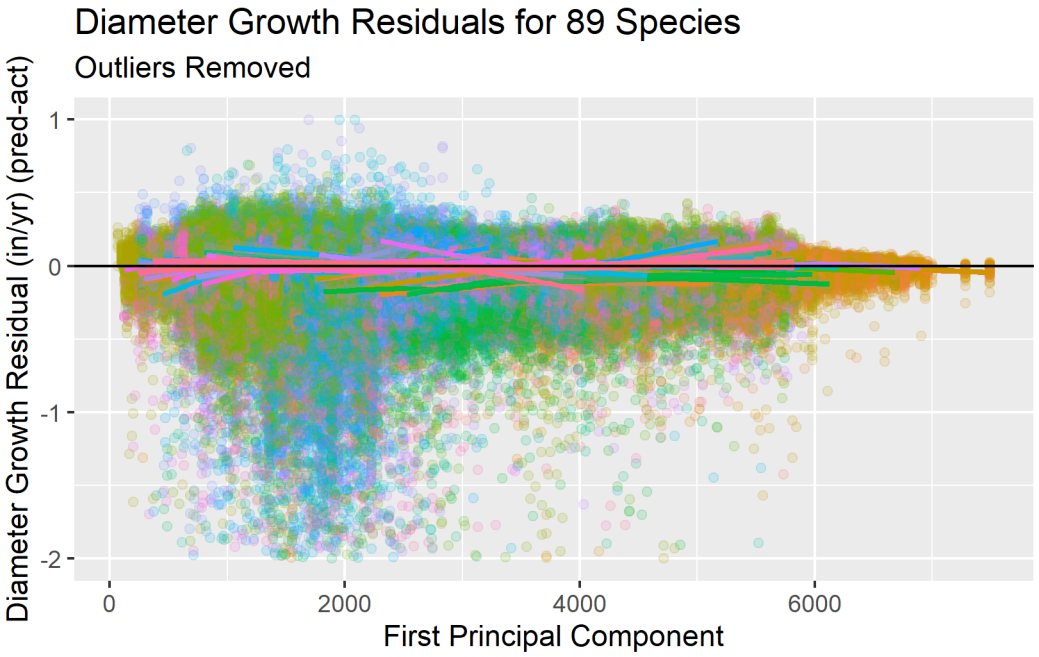


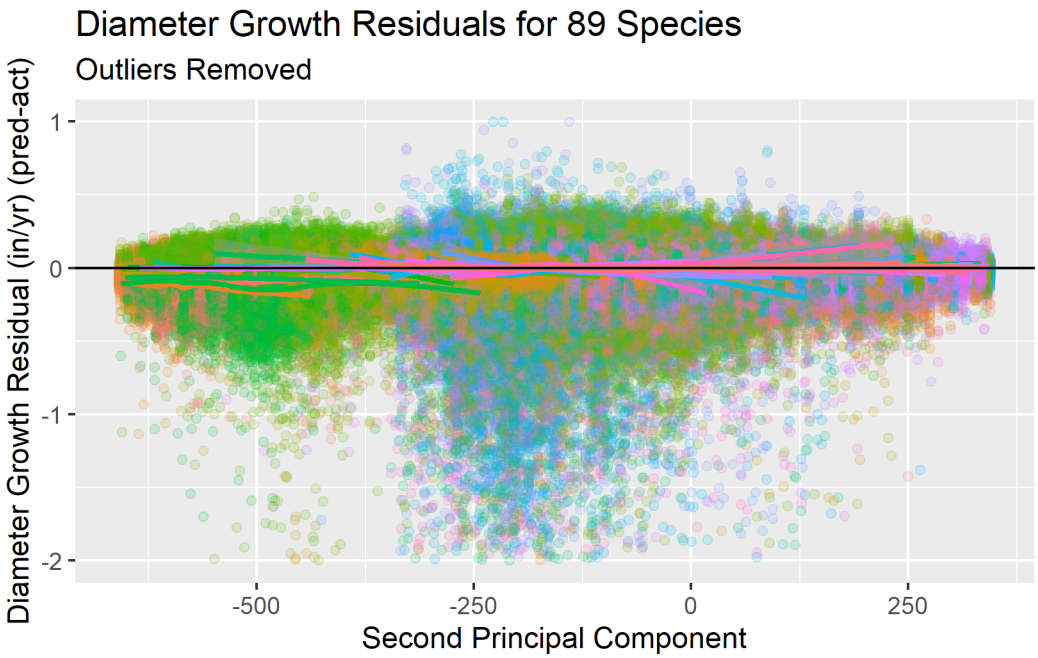
Residual Analysis






Diameter Growth Residuals for 89 Species
Outliers Removed
Diameter Growth Residual (in/yr) (pred-act)
1
0
-1
-2
0.00
0.25
0.50
0.75
1.00
Initial Crown Ratio (fraction of total height)


Diameter Growth Residuals for 89 Species
Outliers Removed
Diameter Growth Residual (in/yr) (pred-act)
1
0
-1
-2
0
2000
4000
6000
First Principal Component


Diameter Growth Residuals for 89 Species
Outliers Removed
Diameter Growth Residual (in/yr) (pred-act)
1
0
-1
-2
-500
-250
0
250
Second Principal Component